Before you commit a large experiment, find out whether it can teach you enough.

EvoForecast is a reproducible digital wind tunnel. It asks a narrower and more useful question than whether evolution is predictable. Can a proposed observation programme tell prospective forecasts apart before a costly physical programme is built?

5,600/5,600
latest blind synthetic SLiM 5.2 runs completed, none failed

0/60
latest forward-looking cells that reached the indicative 50 percent bar

0/120
latest experiment plans classified robustly best

Claim boundary. These results demonstrate synthetic software and experiment-design evidence. Forecast skill in real organisms, qualification of the proposed genome design, and biological validation await empirical testing. They carry no endorsement, partnership, or funding claim.

The choice that sets up every other choice.

Species, generations, populations, starting diversity, conditions, assays, and automation form one coupled design. The wind tunnel tests which combinations could resolve a forecast and which only add observations without telling designs apart.

Claims EF-001 to EF-006 · exact source hashes in audit/source_manifest.json

An auditable chain from parameters to a scored forecast.

REGISTERED PARAMETERS	SLiM 5.2 WORLD	LATENT STATE
OBSERVATION MODEL	MODEL FORECAST	IMMUTABLE DEPOSIT
INDEPENDENT REVEAL	PROPER SCORE	DESIGN / ABSTAIN

Engine direction. SLiM 5.2 is the registered trajectory engine for this round. The challenge harness is engine-agnostic, which means it can work with more than one trajectory engine. Other evolutionary simulators are under evaluation. Later rounds aim to develop ecological network dynamics in a population context and test feedback between ecological change and evolution. This round does not establish that capability.

What the latest challenge says

The genome-plus-phenome track, combining genome data with measured traits, improved its mean continuous ranked probability score, or CRPS, by 47.5 percent over the pedigree baseline, aggregated across 35 registered contexts. CRPS measures forecast accuracy. Its ninety percent prediction range still held the truth only eight times in ten. No forward-looking cell reached the combined bar. Average accuracy, calibration of that range, population-level identifiability, and transfer are separate tests.

Interpretation. This is informative synthetic design evidence. It is not biological forecast skill.

Why controls matter

A corrected favorable marker control produced a 17.0 percent genome-versus-pedigree gain. A matched adverse control produced minus 9.2 percent. The opposite signs show the workflow can expose failure rather than manufacture a genomic advantage.

Supported claim. The software control behaves differently under registered favorable and adverse conditions.

The signature instruments

The build-the-experiment view returns registered gain, coverage, uncertainty, and combined-bar probability for a chosen track, population count, and horizon. The measurement-plan frontier places forecast value against modeled resource use for named packages. Both are precomputed and hash-bound. Browser controls only select rows.

Claims EF-003 to EF-010 and EF-013 · no browser fitting, simulation, or post-outcome optimization

The software is proven here. The biology still needs a real test.

1 · Software proof	SUPPORTED · SYNTHETIC
2 · Indicative design evidence	CONDITIONAL
3 · Empirical qualification	BLOCKED
4 · Operational forecasting	FUTURE

What the theoretical cutaway proposes

A proposed design after the current round would connect diploid ten-chromosome founder variation and sexual recombination to measurable traits, ecology, sampling, laboratory measurements, blind forecasts, and registered scoring. It remains proposed and uncalibrated. The current production model remains clonal.

What remains blocked

Exact founder and genome identity, accepted evidence from the starting and first descendant generations, real measurement performance, material custody, independent prospective reveal, empirical uncertainty analysis, safety authority, and external reproduction.

Run one small round together.

Bring one candidate keystone organism whose effects shape its community, one feasible budget stated in USD, and one outcome that stays hidden until scoring. We rehearse the design in the wind tunnel, then run the smallest real round that can estimate calibration and forecast accuracy under an independent reveal. Later rounds can add a network of interacting species only after that first round clears every gate.

EvoForecast makes a narrower claim. It shows what a real experiment must measure to test evolutionary forecasts, and it reports clearly when the evidence says to wait.

Claims EF-011 to EF-017 · evidence bundle, claim registry, and lineage DAG accompany this note.